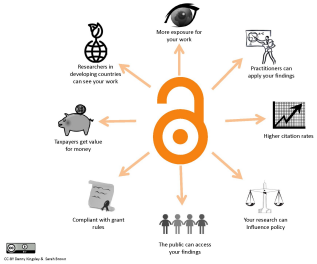
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 7

Data Sharing and Reuse



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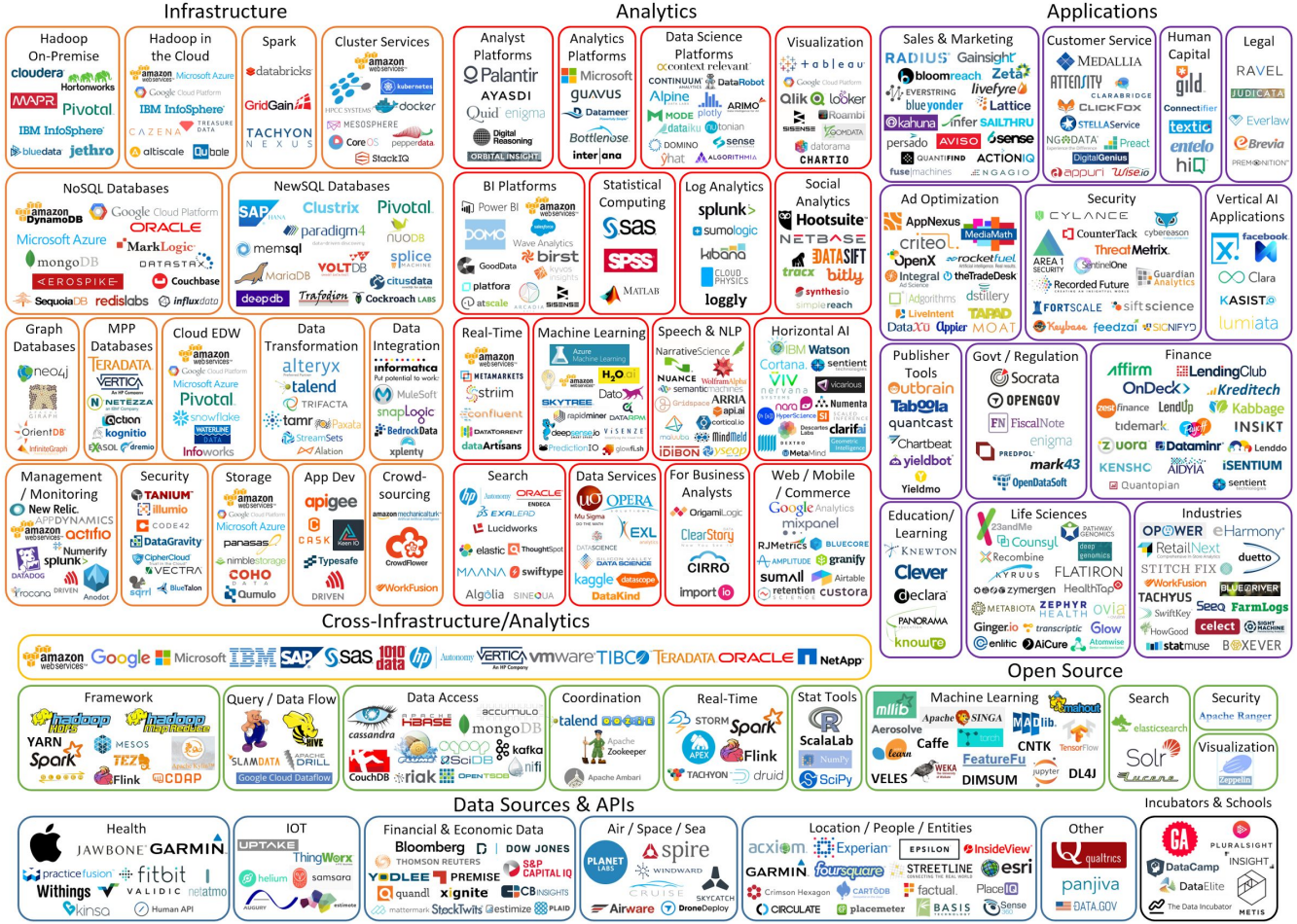
Learning Objectives

1. Describe tools available to foster collaboration and sharing of resources
2. Identify mechanisms to make your software code and/or data citable and sharable
3. Explain distinctions among open access/open science/open data
4. Identify policies for making your data accessible to others
5. Select an appropriate repository for sharing your data
6. Describe measures that can help track the impact of your research data

Big Data Landscape 2016 (Version 3.0)



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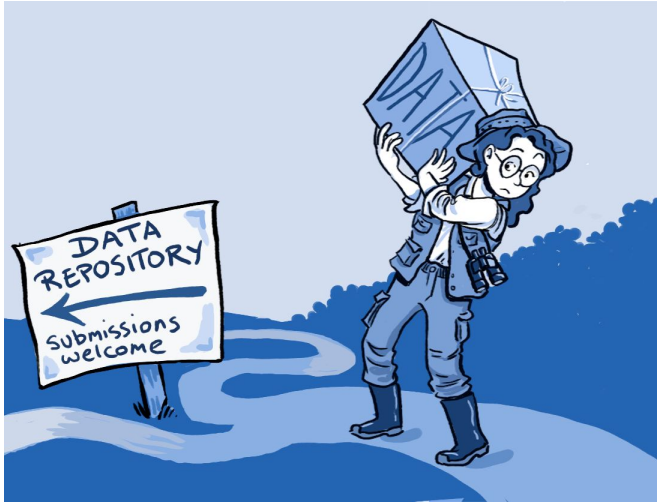
Data Sharing

Data sharing is the practice of making data used for scholarly research available to other investigators

Many funding agencies, institutions, and publication venues have policies regarding data sharing because transparency and openness are considered by many to be part of the scientific method



nature microbiology

ARTICLES

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GRIL-seq provides a method for identifying direct targets of bacterial small regulatory RNA by *in vivo* proximity ligation

Kook Han[1], Brian Tjaden[2] and Stephen Lory[1]*

The first step in the post-transcriptional regulatory function of most bacterial small non-coding RNAs (sRNAs) is base pairing with partially complementary sequences of targeted transcripts. We present a simple method for identifying sRNA targets *in vivo* and defining processing sites of the regulated transcripts. The technique, referred to as global small non-coding RNA target identification by ligation and sequencing (GRIL-seq), is based on preferential ligation of sRNAs to the ends of base-paired targets in bacteria co-expressing T4 RNA ligase, followed by sequencing to identify the chimaeras. In addition to the RNA chaperone Hfq, the GRIL-seq method depends on the activity of the pyrophosphorylase RppH. Using PrrF1, an iron-regulated sRNA in *Pseudomonas aeruginosa*, we demonstrated that direct regulatory targets of this sRNA can readily be identified. Therefore, GRIL-seq represents a powerful tool not only for identifying direct targets of sRNAs in a variety of environments, but also for uncovering novel roles for sRNAs and their targets in complex regulatory networks.

Post-transcriptional regulation mechanisms, primarily through the activities of regulatory small RNAs (sRNAs), play an important role in the bacterial stress response, metabolism, quorum sensing and virulence[1–4]. The sharp increase in the description of this abundant class of RNA regulators during the past decade is the direct result of transcriptome analyses using next-generation RNA sequencing (RNA-seq) and the development of *in silico* methods to identify non-protein-coding transcripts[5]. In bacteria, regulation of gene expression by sRNAs can be divided into two mechanistically distinct categories. One class of sRNAs functions by modifying the activities of regulatory proteins[6]. Other sRNAs regulate gene expression by base pairing with messenger RNAs (mRNAs); this process is accelerated by the RNA chaperone Hfq[7]. The targets of base-pairing sRNAs can range from a few to as many as 1% of total cellular transcripts[8]. Transcripts positively controlled by these sRNAs form secondary structures near their ribosome binding sites and can be disrupted by alternative base pairing with sRNAs, allowing translational initiation[9–11]. Negative regulation depends on base pairing of sRNAs near the translation initiation regions, and on downstream protein coding regions, which can lead to degradation of the transcripts[12].

In spite of the relative ease of identifying regulatory RNAs, their targets are less well defined. The main difficulty in predicting sRNA targets is the limited and non-contiguous base-pairing regions with frequent internal secondary structures and the existence of multiple targets with different base-pairing configurations. Consequently, although several different computational algorithms have been developed, their performance in predicting direct regulatory targets of sRNAs is highly variable[5]. A number of experimental approaches have also been developed to facilitate the identification of direct targets of regulatory RNAs. Several methods (such as cross-linking ligation and sequencing of hybrids (CLASH)[13] and *in vivo* photoactivatable ribonucleoside-enhanced crosslinking and immunoprecipitation (PAR-CLIP)[14]) have been used to identify targets of eukaryotic microRNAs (miRNAs) based on immunoprecipitation of transcripts crosslinked to Argonaut proteins. Crosslinking followed by *in vitro* ligation and sequencing has been used to define the targets of non-coding RNAs in human cells[15]. In bacteria, transcriptome analysis following brief expression of sRNAs can be used to predict likely targets using translational reporters or ribosome profiling[16,17]. However, these methods are often unable to distinguish between direct and indirect effects of sRNA regulation. A variation of these methods was applied to the identification of Hfq-bound mRNAs and sRNAs in *Escherichia coli* and *Salmonella*; however, the assignment of a direct regulatory relationship could not be made[18,19]. Another approach relies on fusing the MS2 coat protein-binding hairpin sequence to the sRNA and capturing the sRNA/mRNA complex with the MS2 bacteriophage coat protein[20].

To facilitate the analysis of global effects of bacterial sRNA–target mRNA interactions, we have developed a robust yet simple method for identifying targets of sRNAs. We have named this method global sRNA target identification by ligation and sequencing (GRIL-seq). The method takes advantage of the proximity of the sRNA and mRNA target sites in a complex that is likely to be stabilized by the Hfq protein. This arrangement facilitates a preferential ligation of the 3′ and 5′ ends by bacteriophage T4 RNA ligase, overexpressed in the same cell, and detection of the chimeric RNAs by sequencing. The GRIL-seq method is an easy, readily accessible approach towards defining post-transcriptional regulatory networks controlled by sRNAs. Conceivably, this method could also be applicable to the analysis of miRNA-directed silencing of eukaryotic mRNAs.

Results

sRNA–target RNA ligation by T4 RNA ligase in the cell. We exploited the ability of bacteriophage T4 RNA ligase to link two base-paired RNA molecules expressed in the same cell. To ligate two RNAs, the 5′ terminal donor sequence must be monophosphorylated. While most bacterial sRNAs are primary transcripts with 5′ triphosphoryl termini, a recent study demonstrated that a fraction of *Salmonella* sRNAs carries 5′ monophosphate, and their enrichment

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The eluted RNA (in 47 μl) was collected, and nuclease-free water (150 μl) was added to reach 200 μl. The enriched RNA was precipitated overnight at −80 °C with 500 μl 100% ethanol (2.5 volumes), 20 μl 0.5 M sodium acetate (0.1 volume) and 4 μl glycogen (1 mg μl^{-1}).

Preparation of the cDNA library for RNA-seq. The precipitated enriched chimeric sRNA was recovered in 15 μl nuclease-free water. The enriched RNAs (100 ng) were used for cDNA library preparation. Library construction for Illumina sequencing was carried out using an NEB Next Ultra Directional RNA Library Prep kit for Illumina (New England Biolabs) following the manufacturer's instructions (Supplementary Notes 1). The index primers used in this study are listed in Supplementary Table 1. To enrich the libraries, 16 cycles of PCR were carried out at the last enrichment step. To obtain high purity of the library from the adapter dimer, band purification was carried out twice at the last step.

Detection of sRNA–target chimaeras by RT–PCR. To remove residual DNA from total RNAs, 10 μg RNA sample in a 50 μl reaction volume was treated with TURBO DNase (4 U) for 30 min, followed by re-inactivation for 5 min using a TURBO DNA-free kit (Thermo Fisher Scientific). Total RNA was recovered (95 μl) by centrifuge at 16,000 *g* for 1 min, and total RNA (5 μg) was converted to cDNA using a SuperScript III First-Strand Synthesis system (Invitrogen) with random hexamers. When required, 1 pmol gene-specific primer was used for the gene-specific cDNA synthesis. Reverse transcription was carried out at 50 °C for 1 h and terminated at 85 °C for 10 min. The residual RNA was removed using 2 μl of an enzyme mixture containing RNase H (2.5 U; New England Biolabs) and RNaseIcocktail (0.5 U; Ambion) for each reaction. Approximately 10% of the reaction was used as the template for PCR amplification using a GoTaq Green Master Mix (Promega) and the primer pairs listed in Supplementary Table 1. Cycling conditions were 95 °C for 3 min, 35 cycles of 94 °C for 20 s, 54 °C for 25 s and 72 °C for 60 s and a final cycle of 72 °C for 5 min. The PCR products were separated by 2% agarose gel electrophoresis, the DNA bands were eluted and cloned into pBTL2 vector (Thermo Fisher Scientific) following the manufacturer's instructions, and the inserts were sequenced using the pBTL2 reverse primer.

RNA-seq data analysis. Sequencing reads were aligned with the TAIR genome using the Bowtie2 aligner[21]. The two biological replicate RNA-seq experiments corresponding to the PNA3pro0 (Up-0) strain resulted in 7,120,964 sequencing reads and 6,693,233 sequencing reads, of which 87% mapped to the reference genome. The two biological replicate RNA-seq experiments corresponding to PNA3 overexpression resulted in 5,687,213 sequencing reads and 5,084,849 sequencing reads, of which 88% mapped to the reference genome. Sequencing read data were normalized using upper quartile normalization[22]. Differential expression in the two conditions was tested using the approach of DESeq[23], where *P* values were computed to indicate the probability of observing such genes's mean expression levels in the two conditions by chance. Because multiple tests were performed across the set of genes, *P* values were corrected to q values to control the false discovery rate at less than 1%[24].

Data availability. The sequencing data have been deposited in the National Center for Biotechnology Information Sequence Read Archive under accession codes SAMN09997141–SAMN09997148.

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Data availability. The sequencing data have been deposited in the National Center for Biotechnology Information Sequence Read Archive under accession codes SAMN05933141–SAMN05933146.

Received 26 November 2015; accepted 25 October 2016; published 22 December 2016

References

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SRX2255953: GRIL-Seq of pseudomonas aeruginosa: replicate 1
1 ILLUMINA (Illumina HiSeq 2500) run: 47.2M spots, 14.3G bases, 8.6Gb downloads

Submitted by: Wellesley College

Study: GRIL-Seq: a method for identifying direct targets of bacterial small regulatory RNA by in vivo proximity ligation
PRJNA349811 • SRP091907 • All experiments • All runs
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Sample:
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Organism: Pseudomonas aeruginosa PAO1

Library:
Name: GS_R1
Instrument: Illumina HiSeq 2500
Strategy: OTHER
Source: TRANSCRIPTOMIC
Selection: RANDOM
Layout: PAIRED

Runs: 1 run, 47.2M spots, 14.3G bases, 8.6Gb

Run	# of Spots	# of Bases	Size	Published
SRR4436850	47,247,376	14.3G	8.6Gb	2016-10-26

ID: 3311130

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SRR4436850	47.2M	14.3Gbp	9.2G	54.2%	2016-10-26	public

This run has 2 reads per spot:

L=151, 100%	L=151, 100%

Legend

Experiment	Library					
SRX2255953	Name	Platform	Strategy	Source	Selection	Layout
to BLAST	GS_R1	Illumina	OTHER	TRANSCRIPTOMIC	RANDOM	PAIRED

Biosample	Sample Description	Organism	Links
SAMN05933141 (SRS1755301)		Pseudomonas aeruginosa PAO1	PRJNA349811 [undefined]

Bioproject	SRA Study	Title
PRJNA349811	SRP091907	GRIL-Seq: a method for identifying direct targets of bacterial small regulatory RNA by in vivo proximity ligation

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Microbe sample from Pseudomonas aeruginosa PAO1

Identifiers BioSample: **SAMN05933141**; Sample name: GRIL-Seq PAO1 Rep1; SRA: SRS1755301

Organism Pseudomonas aeruginosa PAO1
cellular organisms; Bacteria; Proteobacteria; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas; Pseudomonas aeruginosa group; Pseudomonas aeruginosa

Package Microbe; version 1.0

Attributes

strain	PAO1
host	not applicable
collection date	2015
geographic location	USA
sample type	cell culture
replicate	biological replicate 1

BioProject PRJNA349811
Retrieve all samples from this project

Submission Wellesley College, Brian Tjaden; 2016-10-21

Accession: SAMN05933141 ID: 5933141
BioProject SRA

Related information
BioProject
SRA
Taxonomy

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SAMN05933141[accn] (1) BioSample
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"microbe 1 0"[filter] (81189) BioSample
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GRIL-Seq: a method for identifying direct targets of bacterial small regulatory RNA by in vivo proximity ligation

Accession: PRJNA349811 ID: 349811

The first step in the post-transcriptional regulatory function of most bacterial small non-coding RNAs (sRNAs) is base-pairing with partially complementary sequences of targeted transcripts. More...

Accession	PRJNA349811
Data Type	Raw sequence reads
Scope	Multispecies
Grants	• "System and Methods for Analysis of Bacterial Transcriptomes" (Grant ID R15 GM102755, National Institute of General Medical Sciences) • "P. aeruginosa Virulence Determinants" (Grant ID R37 AI021451, National Institute of Allergy and Infectious Diseases Extramural Activities)
Submission	Registration date: 21-Oct-2016 **Wellesley College**

Project Data:

Resource Name	Number of Links
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SRA Experiments	6
OTHER DATABASE	
BioSample	6

SRA Data Details

Parameter	Value
Data volume, Gbases	30
Data volume, Mbytes	18747

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Microbe sample from Pseudomonas aeruginosa PAO1 biosample

SAMN05933141[accn] (1) BioSample

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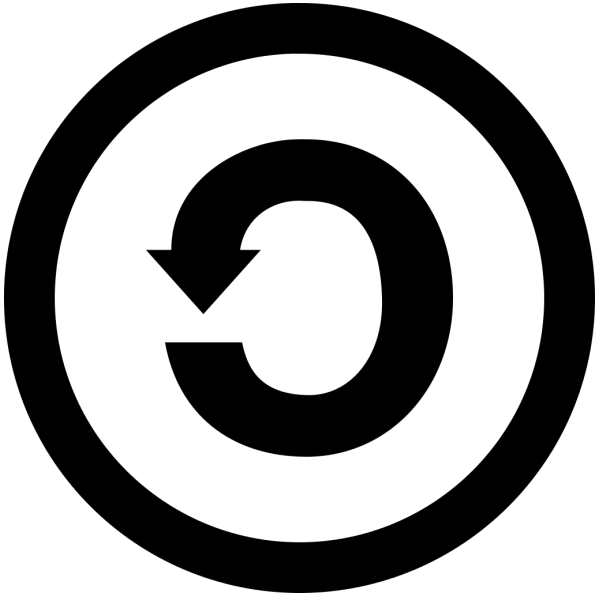

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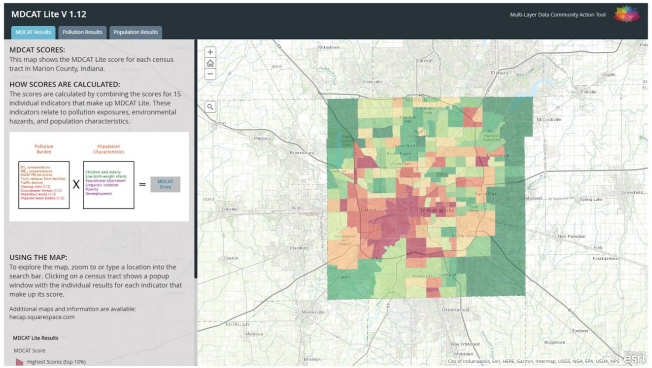

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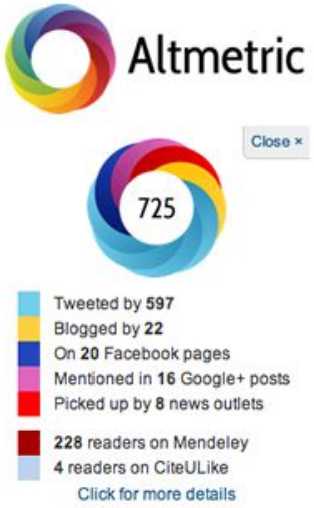

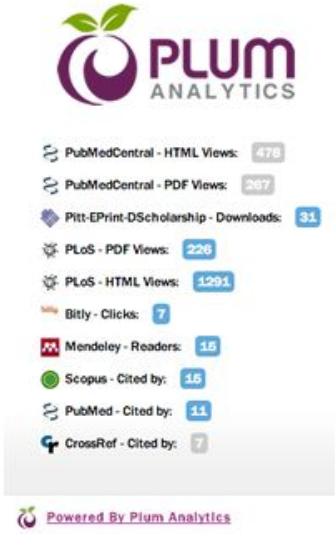

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Kook Han, Brian Tjaden & Stephen Lory

Nature Microbiology **2**,
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doi:10.1038/nmicrobiol.2016.239
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Bacterial physiology
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Received: 26 November 2015
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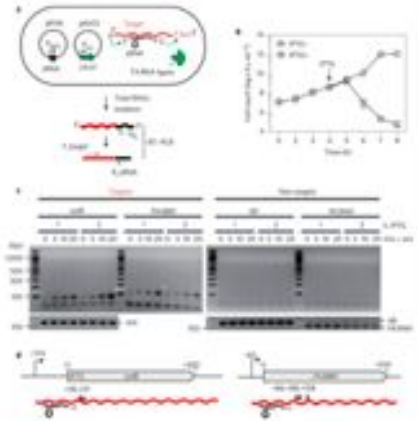
Figure 1: T4 RNA ligase-catalysed *in vivo* linking of sRNA to mRNAs.



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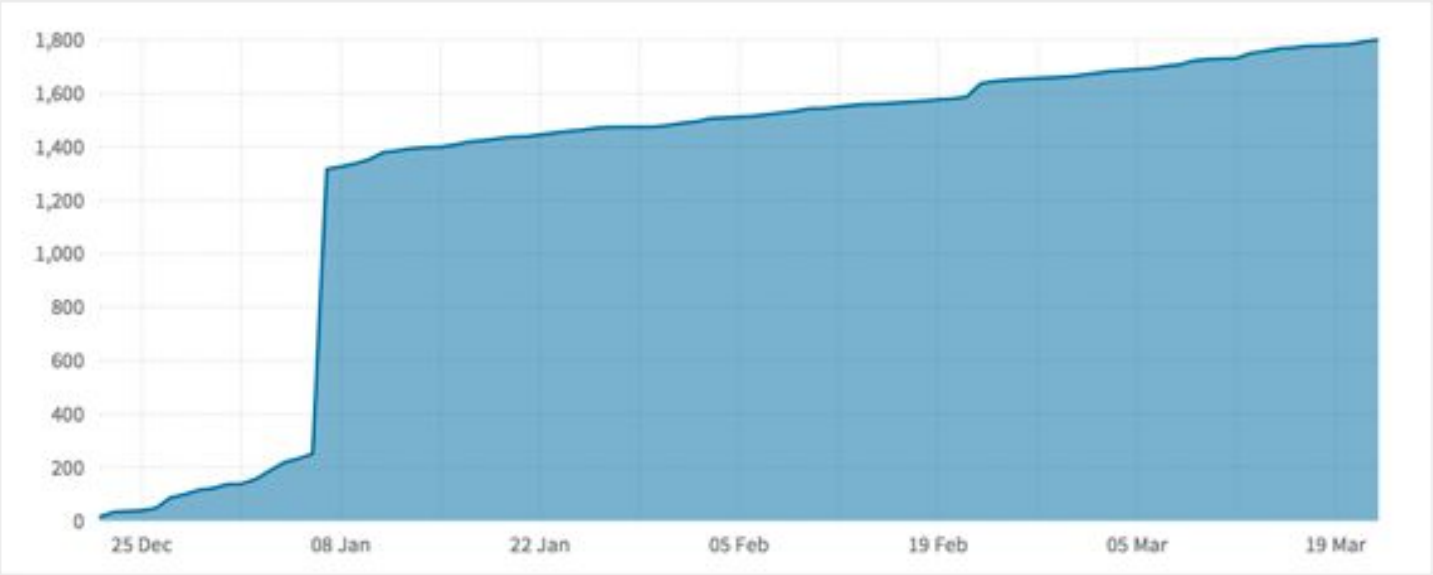

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Nature Microbiology, vol. 16239, issue December (2016) Published by Nature Publishing Group

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K Han, B Tjaden, S Lory - Nature Microbiology, 2016 - nature.com

Abstract The first step in the post-transcriptional regulatory function of most bacterial small non-coding RNAs (sRNAs) is base pairing with partially complementary sequences of targeted transcripts. We present a simple method for identifying sRNA targets in vivo and defining processing sites of the regulated transcripts. The technique, referred to as global small non-coding RNA target identification by ligation and sequencing (GRIL-seq), is ...

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Global snapshots of bacterial RNA networks.

Hör J[1], Vogel J[1,2].

Author information

PMID: 28031253 PMCID: PMC5286387 [Available on 2018-02-01] DOI: 10.15252/embj.201696072

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